Project Charter: Data-Driven Media Optimization

Executive Summary

Project Name: Data-Driven Media Optimization Platform
Client: E3 Media Crafters
Duration: 8 Weeks
Project Manager: Ajit tripathi
Start Date:15 May 2025
End Date:15 July 2025

Business Objective

E3 Media Crafters seeks to leverage artificial intelligence and data analytics to optimize their media strategies and enhance customer engagement through data-driven insights.

Project Goals

Primary Objectives

1. **Sentiment Analysis**: Develop NLP models to analyze customer sentiment from reviews and feedback
2. **Trend Forecasting**: Create time series models to predict media performance and customer behavior
3. **Visual Content Analysis**: Build computer vision models for image classification and content optimization
4. **Interactive Dashboards**: Deploy user-friendly interfaces for stakeholders to access insights

Success Metrics

- **NLP Model**: Achieve 85%+ accuracy in sentiment classification
- **TSA Model**: Forecast accuracy within 15% MAPE
- **CV Model**: Image classification accuracy of 80%+
- **Dashboard**: Real-time data visualization with <3 second load times
- **Deployment**: Fully functional Streamlit application

Scope

In Scope

- Analysis of Yelp dataset (reviews, business data, photos)
- Development of three AI models (NLP, TSA, CV)
- Power BI dashboard creation
- Streamlit web application
- Comprehensive documentation and reports
- Model evaluation and performance analysis

Out of Scope

- Real-time data streaming implementation
- Production-level deployment infrastructure
- Integration with client's existing systems
- Advanced recommendation systems

Stakeholders

Role	Name	Responsibilities
Project Sponsor	E3 Media Crafters	Provide requirements, feedback, and resources
Project Manager	Ajit Tripathi	Overall project coordination and delivery
Data Scientists	Aman Upadhyay	Model development and analysis
Technical Lead	Jatan	Architecture and technical decisions

Timeline & Milestones

Week 1: Business Understanding

- **Deliverables**: Project charter, stakeholder requirements
- **Key Activities**: Define objectives, success criteria, and constraints

Week 2: Data Understanding

- **Deliverables**: Data dictionary, EDA report
- **Key Activities**: Explore Yelp dataset, identify data quality issues

Week 3-4: Data Preparation

- **Deliverables**: Cleaned datasets for modeling
- **Key Activities**: Data preprocessing, feature engineering, data validation

Week 5: NLP Model Development

- **Deliverables**: Sentiment analysis model, evaluation report
- **Key Activities**: Model training, hyperparameter tuning, validation

Week 6: Time Series Analysis

- **Deliverables**: Forecasting model, trend analysis
- **Key Activities**: Time series modeling, seasonal decomposition

Week 7: Computer Vision Model

- **Deliverables**: Image classification model, performance metrics
- **Key Activities**: CNN development, image preprocessing, model evaluation

Week 8: Evaluation & Deployment

- **Deliverables**: Power BI dashboard, Streamlit app, final report
- **Key Activities**: Integration, testing, documentation, presentation

Resources Required

Technical Resources

- Development environment (Python, Jupyter, Git)
- Cloud computing resources (optional)
- Power BI Desktop license
- Yelp dataset access

Human Resources

- 1 Data Scientists (40 hours/week)
- 1 Project Manager (20 hours/week)
- Subject Matter Expert access (5 hours/week)

Risk Assessment

Risk	Probabiliy	Impact	Mitigation Strategy
Data quality issues	Medium	High	Implement robust data validation and cleaning
Model performance below targets	Medium	High	Use ensemble methods and extensive hyperparameter tuning
Timeline delays	Low	Medium	Buffer time built into schedule, agile methodology
Technical challenges	Medium	Medium	Regular technical reviews and expert consultation

Budget Considerations

Estimated Costs

- Development time: 320 hours
- Cloud resources: $200-500
- Software licenses: $100-200
- Total estimated cost: $2,000-3,000

Communication Plan

- **Weekly Status Reports**: Every Friday
- **Stakeholder Updates**: Bi-weekly
- **Technical Reviews**: Weekly
- **Final Presentation**: Week 8

Success Criteria

Technical Success

- All models meet performance benchmarks
- Successful deployment of interactive applications
- Comprehensive documentation delivered

Business Success

- Stakeholder satisfaction score >8/10
- Actionable insights generated for E3 Media Crafters
- Reusable framework for future projects

Approval

Project Sponsor: _____ Date: _____
Project Manager: _____ Date: _____
Technical Lead: _____ Date: _____

This document serves as the foundation for the Data-Driven Media Optimization project and will be referenced throughout the project lifecycle.